

Microstructure-Based Fair Value and Intraday Statistical Arbitrage in XLK

Final report: negative tradable result under incomplete-basket implementation

The market-neutral sparse microprice-signal strategy fails after an exact top-of-book bid/ask boundary-cost audit. The old XLK-ORCL pair-trading diagnostic is also not robust after consistent execution accounting. A separate XLK-only timing extension using the same sparse-basket premium signal remains profitable in Jan, Feb, and March under the bid/ask boundary-cost audit; a stricter stability-island timing rule also remains positive in March.

Selected Holdings

symbol	official_weight_pct	basket_weight
NVDA	15.4537	0.2886
AAPL	12.6167	0.2356
MSFT	9.7708	0.1825
AVGO	5.9634	0.1114
AMD	3.4364	0.0642
CSCO	2.6566	0.0496
ORCL	2.3083	0.0431
CRM	1.3379	0.0250

Minute Data Diagnostics

symbol	minutes	median_spread_bps	trade_count
AAPL	20861	2.97	35436759
AMD	19553	9.33	24537293
AVGO	18860	10.19	22154477
CRM	18530	12.50	8300230
CSCO	22670	2.56	11425050
MSFT	20797	5.15	41606645
NVDA	22693	1.19	138130510
ORCL	18190	7.63	18803754
XLK	23748	0.73	15447039

Sparse Candidate Ranking

subset	train_adf_p	train_half_life_minutes	train_avg_oneway_cost_bps
MSFT NVDA ORCL CRM AMD	0.0149	187.7701	4.4528
NVDA ORCL CRM AMD	0.0221	225.3119	4.1037
NVDA AVGO ORCL CRM AMD	0.0295	196.3795	4.6313
MSFT NVDA AVGO ORCL AMD	0.0471	198.3262	4.5497
NVDA AVGO ORCL AMD	0.0565	233.8761	4.1845

Final Backtest

strategy	sample	trades	gross_bps	cost_bps	net_bps
sparse_e3.5_x1_plain	test	2.000	-82.083	6.950	-89.033
sparse_e3.5_x1_plain	train	6.000	117.726	30.797	86.930
literature_no_trade	test	0.000	0.000	0.000	0.000
literature_no_trade	train	0.000	0.000	0.000	0.000

Sparse Bid/Ask Execution Audit

sample	old_net_bps	new_bidask_cost_bps	new_bidask_net_bps	turnover
test	-89.033	33.159	-115.242	2.000
train	86.930	158.464	-40.737	6.000

Pair Execution Audit

pair	validation_net_bps	test_gross_bps	test_cost_bps	test_net_bps
XLK-ORCL	264.184	509.660	734.756	-225.095
XLK-MSFT	234.674	415.199	648.192	-232.994
XLK-AAPL	279.655	358.664	667.117	-308.453
XLK-CRM	235.937	318.011	633.141	-315.130
XLK-AVGO	104.117	753.636	1153.155	-399.520
XLK-NVDA	337.879	1303.734	1787.467	-483.733
XLK-AMD	266.841	908.588	1453.112	-544.524
XLK-CSCO	220.434	844.951	1485.585	-640.634

Model Comparison

model	train_net_bps	march_net_bps	full_net_bps
baseline_original_best			-579.651
v2_best_by_train_overall	-48.200	-47.636	-95.835
v2_best_march_diagnostic	-118.168	54.077	-64.091
hybrid_sparse_micro_signal_mid_exec	86.930	-89.033	-2.103
literature_no_trade	0.000	0.000	0.000

Selection Audit

model	train_net_bps	march_net_bps	bias_flag
v2_best_by_train_overall	-48.200	-47.636	No test leakage.
v2_best_march_diagnostic	-118.168	54.077	Selection-on-test/look-ahead bias.
hybrid_sparse_micro_signal_mid_exec	86.930	-89.033	No test leakage in selection.
literature_no_trade	0.000	0.000	None.

Positive Timing Extension

period	gross_bps	cost_bps	net_bps	trades	xlk_buyhold_bps
jan	287.185	142.260	144.925	44	-345.091
feb	402.055	137.450	264.605	50	-204.096
mar	870.743	232.390	638.353	58	-370.728
all	1559.983	512.099	1047.883	152	-919.915

Timing Bid/Ask Execution Audit

sample	old_net_bps	new_bidask_cost_bps	new_bidask_net_bps	turnover
feb	264.605	137.453	264.601	50.000
jan	144.925	142.293	144.892	44.000
mar	638.353	232.321	638.422	58.000

Timing Controls

control	train_net_bps	mar_net_bps	avg_pos_mar
selected	409.531	638.353	0.289
sign_flip	-968.949	-1103.133	-0.289
active_always_long	-943.525	-1.391	0.757

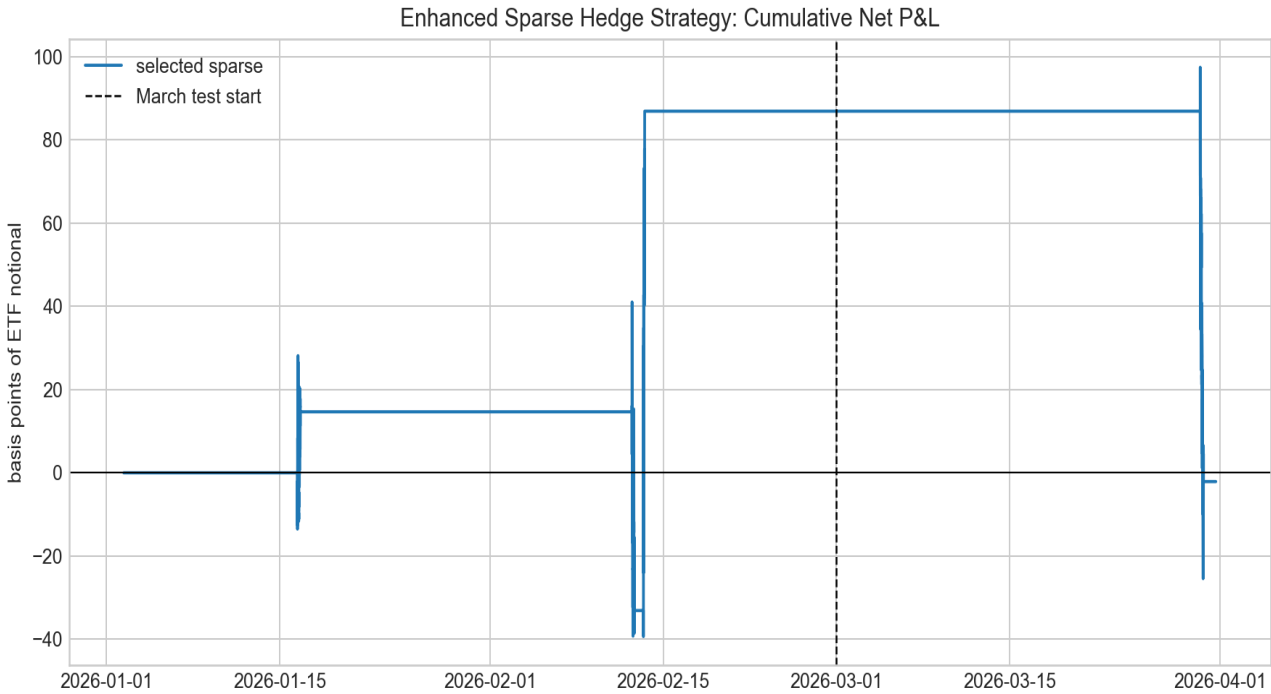
control	train_net_bps	mar_net_bps	avg_pos_mar
active_always_short	384.107	-463.389	-0.757
circular_shift_mean		-440.230	
circular_shift_p95		447.052	
selected_vs_circular_pvalue		0.018	

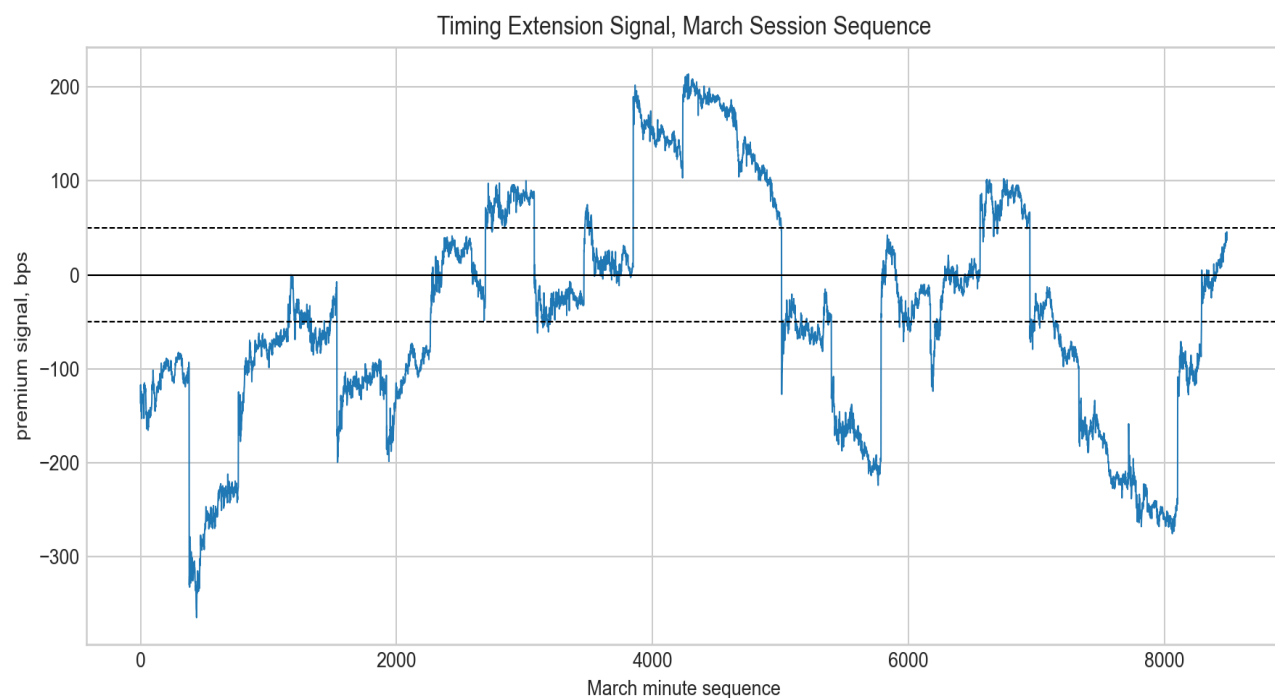
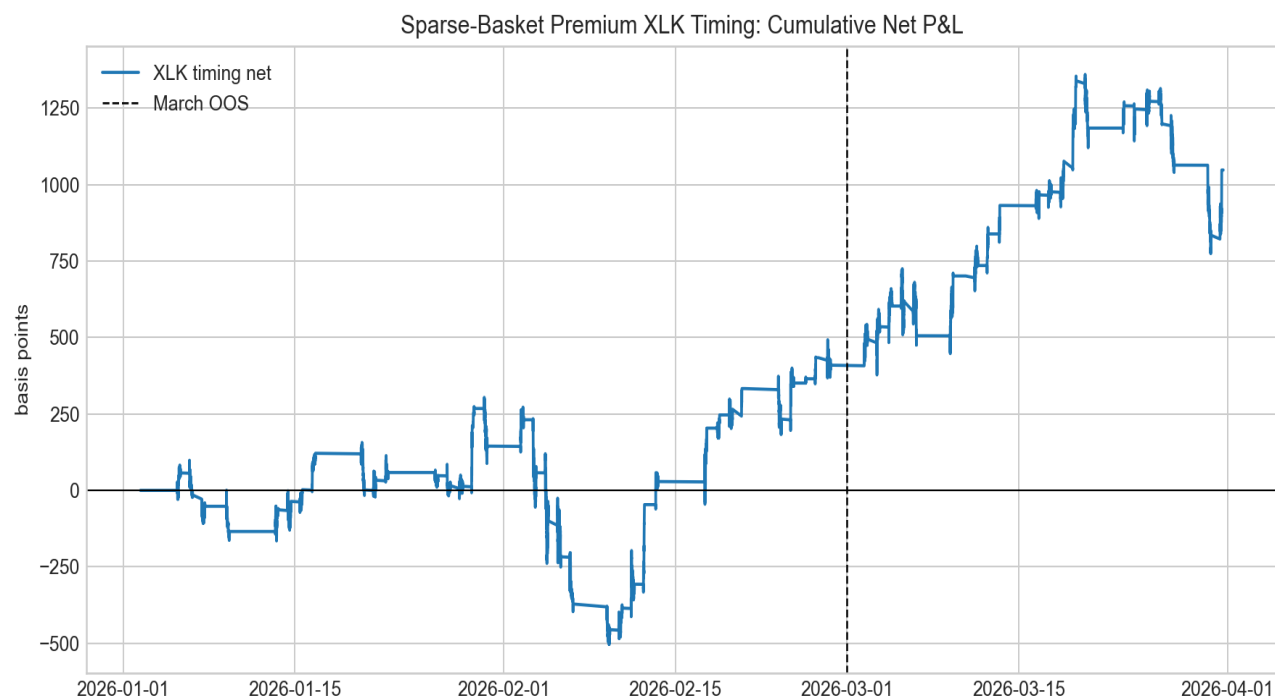
Timing Robustness Decision

decision	selected_strategy	train_net_bps	mar_net_bps	all_net_bps
active_xlk_timing	micro_shrink_0.75_cw10d_e60_x0_mh240	1025.795	120.050	1145.846

Timing Stability Islands

signal_view	center_days	entry_bps	valid_rules	median_train_net_bps	median_mar_net_bps
micro_shrink_0.75	10	60.000	5	731.365	71.233
micro_shrink_0.50	10	60.000	6	606.186	62.057
micro_shrink_0.25	10	60.000	6	574.601	120.050
micro	10	30.000	1	1097.222	-0.913
micro	10	60.000	5	652.846	75.721





Conclusion

The proposed market-neutral ETF-basket arbitrage fails under the available eight-name proxy basket. Microprice is nevertheless useful as a fair-value signal: when the noisy hedge leg is removed, the sparse-basket premium produces a

profitable XLK-only timing strategy in this sample.